

Statistics: Preliminary Ph.D. Course

Day 2: Random Variables

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Today's Session

We will more or less follow the structure in Hansen's Book

- Univariate **Discrete Random Variables**. Definition and Examples. Support and PMFs. Expectation Operator and Linearity. Definitions, Theorems, Proofs, and Examples.
- Univariate **Continuous Random Variables**. Support, PDFs, and Transformations. Expectation Operator and Linearity. Definitions, Theorems, Proofs, and Examples.
- The **Distribution Function** (CDF): Definition and Properties. Relation to Axioms of Probability.
- **Moments** and the Variance Operator: Definition, Properties, and Examples.
- **Jensen's Inequality**: Theorem, Proof, and Examples.
- **Symmetric Distributions**: Definition, Propositions, Proofs, and Examples.
- **MGF** and **ChF**: Definition, Theorems, Proofs, and Examples.
- **Generalization** to the Bivariate and Multivariate cases. Covariances.

Random Variables

Definition

- It is convenient to represent random outcomes numerically. *Example.*
- **Definition 2.1:** A (univariate) **Random Variable** X is a real-valued outcome, i.e., a function from the sample space Ω to the real line $\mathbb{R}$:

$$X(\omega) : \Omega \longrightarrow \mathbb{R}$$

$\omega \mapsto X$

- Some random variables take values only on subsets of the real line. *Examples.*
- **Definition 2.2:** A set $\mathcal{X}$ is **Discrete** if it has a finite or countably infinite number of elements. *Examples.*
- **Definition 2.3:** If there is a discrete set $\mathcal{X}$ such that $\mathbb{P}(X \in \mathcal{X}) = 1$, then X is a **Discrete Random Variable**.

Discrete Case: Support, PMF, and Expectation

Definitions

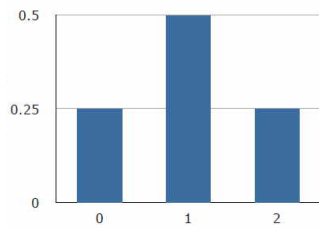
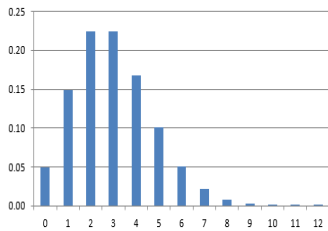
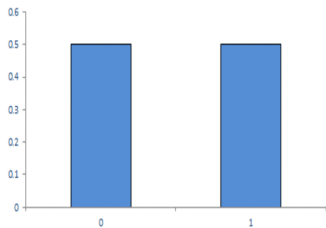
- **Definition 2.4:** The smallest set $\mathcal{X}$ with the property $\mathbb{P}(X \in \mathcal{X}) = 1$ is the **Support** of the random variable X . It is the set of values with a strictly positive probability of occurrence: $\mathcal{X} = \{\tau_1, \tau_2, \dots\}$. We call the points τ_j the **Support Points**.
- **Definition 2.5:** The **Probability Mass Function (PMF)** of a discrete random variable X is $\pi(x) = \mathbb{P}(X = x)$, the probability that X takes the value x . When evaluated at the support points τ_j we write $\pi_j = \pi(\tau_j)$.
- A necessary condition for a valid PMF is $\sum_j \pi_j = 1$. *Examples.*
- **Definition 2.6:** For a discrete random variable X with support points τ_j , the **Expectation** of X is:

$$\mathbb{E}[X] = \sum_j \tau_j \pi(\tau_j)$$

Examples.

Probability Mass Functions

An Illustration



Discrete Case: The Expectations Operator

Properties

- **Definition 2.7:** For a discrete random variable X with support points τ_j , the **Expectation** of the transformation $g(X)$ is:

$$\mathbb{E}[g(X)] = \sum_j g(\tau_j)\pi(\tau_j)$$

Theorem 2.1 (Linearity of Expectations)

For any constants $a, b \in \mathbb{R}$, $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$.

Proof. (Hint: Use the definition of expectation and the basic properties of sums).

- Expectations can be finite, infinite, or undefined, depending on the series $\sum_j$. If it is not convergent, it may be either infinite or undefined.

Examples.

The Distribution Function

- Any random variable (either continuous or discrete) X can be represented by its distribution function: $X \sim F$.
- **Definition 2.8:** The **Distribution Function** is $F(x) = \mathbb{P}(X \leq x)$, i.e., the probability of the event $\{X \leq x\}$. It is also called the **Cumulative Distribution Function (CDF)**.
- For the discrete case, the CDF is: $F(\tau_k) = \mathbb{P}(X \leq \tau_k) = \sum_{j=1}^k \pi(\tau_j)$.

Theorem 2.2 (Properties of a CDF)

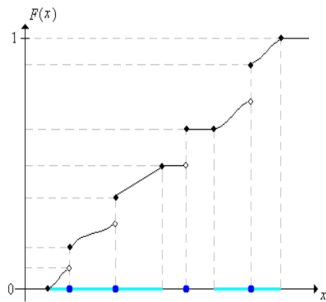
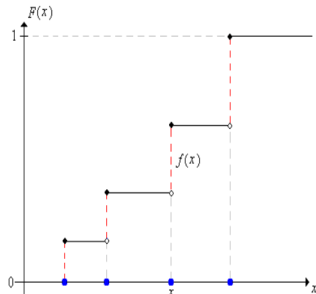
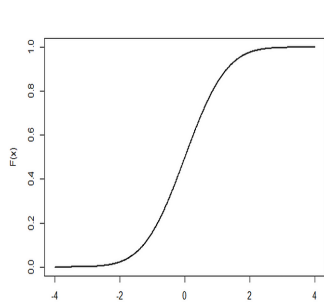
Let $F(x)$ be a distribution function, $F : \mathbb{R} \rightarrow [0, 1]$. Then:

1. $F(x)$ is non-decreasing.
2. $\lim_{x \rightarrow -\infty} F(x) = 0$.
3. $\lim_{x \rightarrow +\infty} F(x) = 1$.
4. $F(x)$ is right-continuous: $\lim_{x \rightarrow x_0^+} F(x) = F(x_0)$.

Relation to Axioms. Cadlag Functions.

The Distribution Function

An Illustration



Continuous Case: PDF, Support, and Expectation

Definitions

- If a random variable X takes a continuum of values it is not discretely distributed.
- **Definition 2.9:** If $X \sim F$, and $F(x)$ is continuous, then X is a **Continuous Random Variable**.
- An **important property** is that the probability that X equals any specific value is zero: $\mathbb{P}(X = x) = 0$.
Proof. (Hint: Take the event $[x, x + \epsilon]$, use the definition of CDF, and take the limit as $\epsilon \rightarrow 0$).

Examples.

- Continuous random variables do not have PMF. An analog is the derivative¹ of F .
- **Definition 2.10:** When $F(x)$ is differentiable, its **Probability Density Function (PDF)** is

$$f(x) = \frac{d}{dx}F(x)$$

¹More on this later, in the Measure Theory part.

Continuous Case: PDF, Support, and Expectation

Definitions

Theorem 2.3 (Properties of a PDF)

A function $f(x)$ is a density function if and only if:

1. $f(x) \geq 0 \ \forall x$.
2. $\int_{\mathbb{R}} f(x) \, dx = 1$.

Examples.

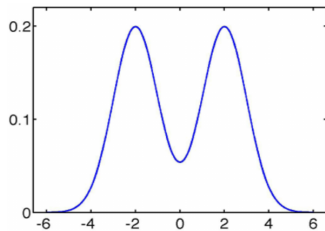
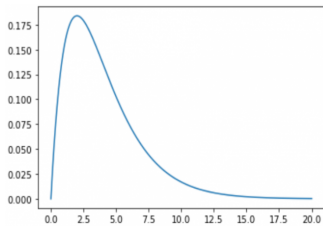
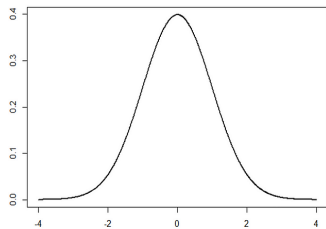
- We can use densities to compute probabilities of events (sets):

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) \, dx$$

- **Definition 2.11:** The **Support** $\mathcal{X}$ of a continuous random variable is the smallest set containing $\{x : f(x) > 0\}$.

Probability Density Functions

An Illustration



Continuous Case: PDF, Support, and Expectation

Definitions

- **Definition 2.12:** If X is continuously distributed with density $f(x)$ its **Expectation** is defined as:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x f(x) dx$$

- **Definition 2.13:** If X has density $f(x)$ then the **Expectation** of $g(X)$ is:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx$$

Examples. Finiteness vs Infiniteness vs Non-existence.

Theorem 2.4 (Linearity of Expectations)

For any constants $a, b \in \mathbb{R}$, $\mathbb{E}[a + bX] = a + b\mathbb{E}[X]$.

Proof. (Hint: Use the definition of expectation and the properties of integrals).

Transformations

Theorem and Applications

A natural question is: If $X \sim F$ for some known F , what is the distribution of $Y = g(X)$?

Theorem 2.5 (Distributions of Monotonic Transformations)

If $X \sim f_X(x)$, $f(x)$ is continuous on $\mathcal{X}$, $g(x)$ is strictly monotone, and $g^{-1}(y)$ is continuously differentiable on $\mathcal{Y}$, then for $y \in \mathcal{Y}$:

$$f_Y(y) = f_X(g^{-1}(y)) |J(y)|$$

where $J(y) = \frac{d}{dy}g^{-1}(y)$.

Applications.

Probability Integral Transformations (PIT).

Examples of Non-Monotonic Transformations.

The Variance Operator

Definitions, Properties, and Examples

- **Definition 2.14:** The **Variance** of X is $\sigma^2 = \text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$.
- The variance is necessarily greater than zero, and either finite or infinite. It is zero only for degenerate random variables.
- **Definition 2.15:** A random variable is **Degenerate** if for some value $c \in \mathbb{R}$, $\mathbb{P}(X = c) = 1$. It is essentially non-random.
- **Definition 2.16:** The **Standard Deviation** of X is the positive square root of the variance: $\sigma = \sqrt{\sigma^2}$.
- It is typical to use the mean and standard deviation to summarize the center and spread of a distribution.

The Variance Operator

Definitions, Properties, and Examples

Theorem 2.6

$$\mathbb{V}ar[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Proof.

Theorem 2.7 (Quadratic Property of the Variance Operator)

For any constants $a, b \in \mathbb{R}$:

$$\mathbb{V}ar[a + bX] = b^2 \mathbb{V}ar[X]$$

Proof. (Hint: Use the definition of variance).

Examples.

Moments of Random Variables

Definition and Examples

- **Definition 2.17:** The m -th moment of a random variable X is $\mathbb{E}[X^m]$.
- For non-negative X , m can be real-valued. Even moments are either finite or undefined. Odd moments may be finite, infinite, or undefined.
- An important result is that if $\mathbb{E}[|X|^r] < \infty$ then $\mathbb{E}[|X|^s] < \infty \forall 0 \leq s \leq r$. (See Lyapunov's Inequality in Hansen's Book pg. 43).

Examples.

Jensen's Inequality

Theorem, Proof, and Examples

Theorem 2.8 (Jensen's Inequality)

For any random variable X , if $g(\cdot)$ is a convex function then:

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$$

If $g(\cdot)$ is a concave function, then:

$$g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$$

Proof (Convex Case). (Hint: Use the tangent line to $g(x)$ at $x = \mathbb{E}[X]$ and take expectations).

Examples.

Applications.

Symmetric Distributions

Definition

- **Definition 2.18:** A random variable has a **Symmetric Distribution** about 0 if the distribution function F satisfies:

$$F(x) = 1 - F(-x)$$

Furthermore, if X has a density $f(x)$:

$$f(x) = f(-x)$$

i.e., if its density is an even function.

- There are some interesting results for symmetric distributions.

Symmetric Distributions

Results

Proposition 2.1 (Odd Moments of Symmetric Distributions)

If a distribution is symmetric about zero then all odd moments are zero if the moment is finite.

Proof. (Hint: Assume that the density exists, use the definition of moment, integrate separately on the positive and negative reals, and use the definition of symmetry).

Theorem 2.9 (Odd Functions and Symmetric Distributions)

If $f(x)$ is symmetric about zero, $g(x)$ is odd, and $\int_{\mathbb{R}_+} |g(x)|f(x)dx < \infty$, then $\mathbb{E}[g(X)] = 0$

Proof. (Hint: Use the definition of expectation, integrate separately on the positive and negative reals, and use the definition of odd function).

The Moment Generating Function

Definition and Theorems

- **Definition 2.19:** The **Moment Generating Function (MGF)** of X is $M(t) = \mathbb{E}[\exp(tX)]$.

Examples.

Theorem 2.10 (Moments and the MGF)

If $M(t)$ is finite for t in a neighborhood of 0 then $M(0) = 1$ and:

$$\left. \frac{d^m}{dt^m} M(t) \right|_{t=0} = \mathbb{E}[X^m]$$

Proof (Continuous Case). (Hint: Use the definition of expectation and assume you can differentiate under the integral sign).

Examples.

The Characteristic Function (ChF)

Definition and Theorems

- Since the MGF may not be finite, the ChF may be used instead for advanced proofs.
- **Definition 2.20:** The **Characteristic Function (ChF)** of X is $Ch(t) = \mathbb{E}[\exp(itX)]$, where $i = \sqrt{-1}$.

Theorem 2.11 (Moments and the ChF)

The characteristic function satisfies: $Ch(0) = 1$ and $\frac{d^m}{dt^m} Ch(t) \big|_{t=0} = i^m \mathbb{E}[X^m]$.

Proof (Continuous Case). (Hint: The same as for the MGF).

Theorem 2.12 (Symmetry and the ChF)

If X is symmetrically distributed about zero its characteristic function equals $Ch(t) = \mathbb{E}[\cos(tX)]$.

Proof. (Hint: Use Euler's Formula $\exp(iu) = \cos(u) + i \sin(u)$ and Theorem 2.9).

Bivariate Random Vectors

Definitions

- We can **generalize** the previous discussion to higher dimensions, $\mathbb{R}^m$. We will focus on the bivariate case $m = 2$.
- **Definition 2.21:** A bivariate **Random Vector** $(X, Y)'$ is a pair of real-valued outcomes, i.e., a function from the sample space Ω to $\mathbb{R}^2$:

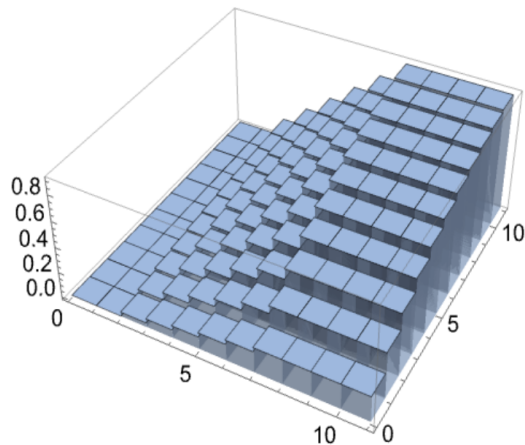
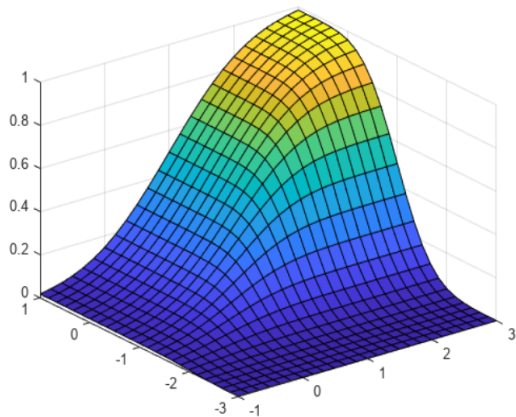
$$\begin{pmatrix} X \\ Y \end{pmatrix} : \Omega_1 \times \Omega_2 \longrightarrow \mathbb{R}^2$$
$$(\omega_1, \omega_2) \mapsto (x, y)$$

- **Definition 2.22:** The **Joint Distribution Function** of the bivariate random vector $(X, Y)'$ is $F(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\})$. (Same properties).

Examples.

Bivariate Random Vectors

Joint Distribution Function: An Illustration



Bivariate Random Vectors

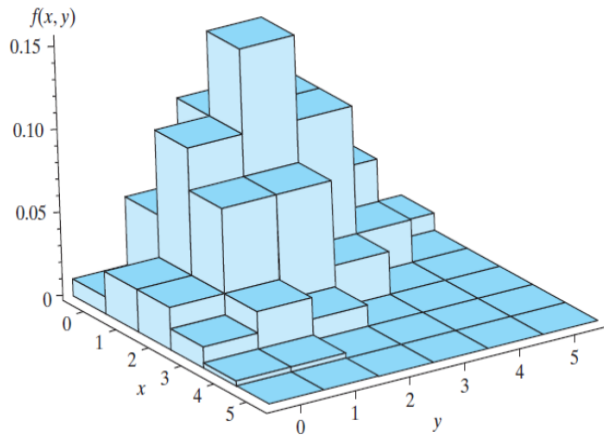
Joint PMFs

- **Definition 2.23:** A pair of random variables is **Discrete** if there is a discrete set $\mathcal{S} \subseteq \mathbb{R}^2$ such that $\mathbb{P}((X, Y)' \in \mathcal{S})$.
- **Definition 2.24:** The set $\mathcal{S} \subseteq \mathbb{R}^2$ is the **Support** of (X, Y) and consists of a set of points in $\mathbb{R}^2$. In many cases $\mathcal{S} = \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subseteq \mathbb{R}$ and $\mathcal{Y} \subseteq \mathbb{R}$. And we can write the **Support Points** as $\{\tau_1^x, \tau_2^x, \dots\}$ and $\{\tau_1^y, \tau_2^y, \dots\}$.
- **Definition 2.25:** The **Joint Probability Mass Function (PMF)** of the bivariate random vector $(X, Y)'$ is $\pi(x, y) = \mathbb{P}(X = x, Y = y)$. At the support points we write $\pi_{ij} = \pi(\tau_i^x, \tau_j^y)$.
- A necessary condition for a valid Joint PMF is $\sum_i \sum_j \pi_{ij} = 1$.

Example.

Bivariate Random Vectors

Joint PMF: An Illustration



Bivariate Random Vectors

Joint PDFs

- The pair $(X, Y)'$ is continuously distributed if its CDF $F(x, y)$ is continuous in (x, y) .
- **Definition 2.26:** When $F(x, y)$ is differentiable, its **Joint Probability Density Function (PDF)**, $f(x, y)$ is

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

They are non-negative functions that integrate to one in $\mathbb{R}^2$ (these are the two conditions for a valid joint density).

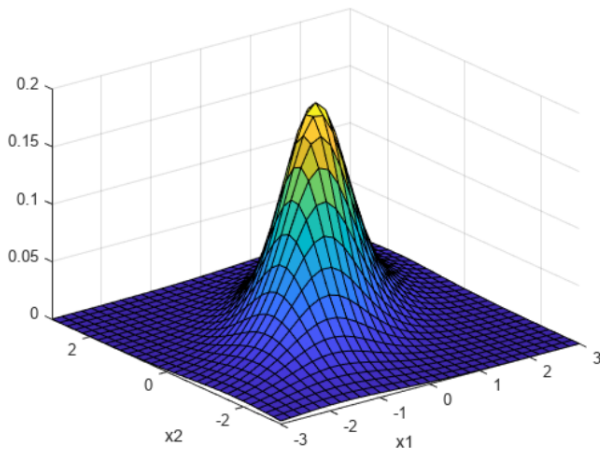
- As in the univariate case, we can use them to compute probabilities:

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y) \, dx dy$$

Examples.

Bivariate Random Vectors

Joint PDF: An Illustration



Bivariate Random Vectors

Marginal Distributions

- The joint distribution of $(X, Y)'$ fully describes the distribution of each component of the random vector.
- **Definition 2.27:** Let $(X, Y)'$ be a bivariate random vector. The Marginal Distribution of X is $F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X \leq x, Y \leq \infty) = \lim_{y \rightarrow +\infty} F(x, y)$.
- In the continuous and discrete cases we write, respectively:

$$F_X(x) = \lim_{y \rightarrow +\infty} \int_{-\infty}^y \int_{-\infty}^x f(x, y) \, dx dy = \int_{\mathbb{R}} \int_{-\infty}^x f(x, y) \, dx dy$$

$$F_X(x) = \lim_{\tau_k \rightarrow +\infty} \sum_{j=1}^{\tau_k} \sum_{i=1}^{\tau_m} \pi_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\tau_m} \pi_{ij} = \sum_j \sum_{i=1}^{\tau_m} \pi_{ij}$$

Examples.

Bivariate Random Vectors

Marginal Densities and Expectations

- **Definition 2.28:** The **Marginal Density** of X and that of Y given a joint density $f(x, y)$ are:

$$f_X(x) = \int_{\mathbb{R}} f(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{\mathbb{R}} f(x, y) dx$$

- **Definition 2.29:** Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then:

$$\mathbb{E}[(X, Y)'] = (\mathbb{E}[X], \mathbb{E}[Y])'$$

$$\mathbb{E}[g(X, Y)] = \int_{\mathbb{R}} \int_{\mathbb{R}} g(x, y) f(x, y) dx dy \quad \text{in the continuous case}$$

$$\sum_{(x, y) \in \mathbb{R}^2 : \pi(x, y) > 0} g(x, y) \pi(x, y) \quad \text{in the discrete case}$$

Examples.

Bivariate Random Vectors

Independence

- Recall the definition of independence: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Consider the events $A = \{X \leq x\}$ and $B = \{Y \leq y\}$. So that $\mathbb{P}(A \cap B) = \mathbb{P}(\{X \leq x\} \cap \{Y \leq y\}) = F(x, y)$.
- **Definition 2.30:** The random variables X and Y are said to be **Statistically Independent**, written $X \perp\!\!\!\perp Y$ or $X \perp Y$, if for all x and y : $F(x, y) = F_X(x)F_Y(y)$.

Theorem 2.13 (Independence with PDFs and PMFs)

The discrete random variables X and Y are independent if $\forall x, y, \pi(x, y) = \pi(x)\pi(y)$. Moreover, if X and Y have a continuous and differentiable joint distribution function, they are statistically independent if $\forall x, y, f(x, y) = f_X(x)f_Y(y)$.

Proof. (Hint: Use the definition of independence and differentiate).

Bivariate Random Vectors

Independence

Theorem 2.14 (Independence and MGFs)

If X and Y are independent with MGFs $M_X(t)$ and $M_Y(t)$, then the MGF of $Z = X + Y$ is $M_Z = M_X(t)M_Y(t)$.

Proof. (Hint: Use the definitions of MGF and Theorem 2.13).

- In fact, we can prove the following:

Theorem 2.15 (Independence and Expectations)

If X and Y are statistically independent then for any two functions $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $\mathbb{E}[|g(X)|] < \infty$ and $\mathbb{E}[|h(Y)|] < \infty$, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$.

Proof. (Hint: Use the definition of independence and Theorem 2.13).

Bivariate Random Vectors

Independence

Theorem 2.16 (Independence and Transformations)

If X and Y are statistically independent then for any two functions $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$, $U = g(X)$ and $V = h(Y)$ are independent.

Proof. (Hint: Use the definition of independence and define the pre-image sets $A(u) = \{x : g(x) \leq u\}$ and $B(v) = \{y : h(y) \leq v\}$).

Examples.

Bivariate Random Vectors

Covariance and Correlation

- **Definition 2.31:** The **Covariance** between X and Y is

$$\mathbb{C}ov(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

- **Definition 2.32:** The **Correlation** between X and Y is

$$\rho(X, Y) = \frac{\mathbb{C}ov(X, Y)}{\sqrt{\mathbb{V}ar[X]}\sqrt{\mathbb{V}ar[Y]}}$$

- If $\mathbb{C}ov(X, Y) = 0$ the $\rho(X, Y) = 0$, and it is typical to say that X and Y are uncorrelated.

Theorem 2.17 (Independence and Correlation)

If two random variables X and Y are statistically independent with finite variances, they are uncorrelated. The reverse is not true.

Proof. (Hint: Use the definitions of covariance and independence, then find a counterexample of two dependent random variables to prove that the reverse is not true. Think about symmetry).

Bivariate Random Vectors

Covariance and Correlation

Theorem 2.18 (Property of the Variance)

If X and Y have finite variances then $\mathbb{V}ar[X \pm Y] = \mathbb{V}ar[X] + \mathbb{V}ar[Y] \pm 2\mathbb{C}ov(X, Y)$.

Corollary 2.18

If X and Y are uncorrelated and have finite variances then $\mathbb{V}ar[X \pm Y] = \mathbb{V}ar[X] + \mathbb{V}ar[Y]$.

Examples.

Multivariate Random Vectors

Key Definitions

- We now consider the case of an $m \times 1$ random vector $X \in \mathbb{R}^m$: $(X_1, \dots, X_m)'$.
- **Definition 2.33:** The **Joint Distribution Function** is $F(x) = \mathbb{P}(X_1 \leq x_1, \dots, X_m \leq x_m)$.
- **Definition 2.34:** The **Joint PMF** is $\pi(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X_1 = x_1, \dots, X_m = x_m)$.
- **Definition 2.35:** When $F(x)$ is differentiable, its **Joint PDF** equals:

$$f(x) = \frac{\partial^m}{\partial x_1 \dots \partial x_m} F(x)$$

- **Definition 2.36:** The **Expectation** of X given by the vector of expectations:

$$\mathbb{E}[X] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_m] \end{pmatrix}$$

Multivariate Random Vectors

Key Definitions

- **Definition 2.37:** The $m \times m$ **Variance-Covariance Matrix** Σ of X is given by

$$\mathbb{V}ar[X] = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])'] = \mathbb{E}[XX'] - \mathbb{E}[X]\mathbb{E}[X]' = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1m} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{m1} & \sigma_{m2} & \cdots & \sigma_m^2 \end{pmatrix}$$

where $\sigma_j^2 = \mathbb{V}ar[X_j]$ and $\sigma_{ij} = \mathbb{C}ov(X_i, X_j)$ for $i, j = 1, \dots, m$ and $i \neq j$.

Theorem 2.19 (Properties of the Var-Cov Matrix)

1. Σ is Symmetric.
2. Σ is Positive Semidefinite (p.s.d.).

Proof. (Hint: Use the definition of symmetric matrices, then show that the quadratic form $a'\Sigma a \geq 0$ for some $a \in \mathbb{R}^m$).

Multivariate Random Vectors

Key Definitions

Theorem 2.20 (Var-Cov Matrix of Linear Combinations)

If the random vector X with $m \times 1$ expectation μ and $m \times m$ variance-covariance matrix Σ and A is a $q \times m$ matrix, then AX is a $q \times 1$ random vector with expectation $A\mu$ and $q \times q$ variance-covariance matrix $A\Sigma A'$.

Proof. (Hint: Use the definitions of expectations and variance-covariance matrix).

Theorem 2.21 (Finiteness of Multivariate Expectations)

For $X \in \mathbb{R}^m$, $\mathbb{E}[\|X\|] < \infty$ if and only if $\mathbb{E}[|X_j|] < \infty$ for $j = 1, \dots, m$.

Proof. (For the implication use the triangle inequality for the vector Euclidean norm. For the 'only if' part of the statement use the fact that the Euclidean norm of a vector is larger than the length of any individual component).

Multivariate Random Vectors

- Finally, let's see a very useful multivariate generalization of Theorem 2.5.

Theorem 2.22 (Distributions of Monotonic Transformations)

Let $X \in \mathbb{R}^m$ have density $f_X(x)$, let g be a one-to-one function $g : \mathbb{R}^m \rightarrow \mathbb{R}^q$, and let $h(y) = g^{-1}(y)$ be differentiable. Then, the random variable $Y = g(X)$ has density:

$$f_Y(y) = f_X(h(y))|\det(J(y))|$$

where $J(y) = \partial h(y)/\partial y'$ is the Jacobian of $h(y)$.

Examples.

Recommended Exercises

Hansen's Book

- Chapter 2, Exercises: 2.1-2.16.
- Chapter 4, Exercises: 4.1a)-c), 4.1 e)-f), 4.2 a)-d), 4.3 a)-c), 4.4-4.12, 4.14, 4.20, 4.22.

References

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Wasserman, L. (2004). All of Statistics: A Concise Course in Statistical Inference. *Springer*.

Shao, J. (1999). Mathematical Statistics. *Springer*.